

1 **Error propagation from velocity uncertainty to AVO attribute misclassification:**
2 **A validated closed-form framework applied to CO₂-storage monitoring at**
3 **Smeaheia**

4 Annaliese J. Lee Seyon¹ and Abdullatif Alshuhail¹

5 ¹Department of Geosciences, College of Petroleum Engineering and Geosciences,
6 King Fahd University of Petroleum and Minerals (KFUPM), Dhahran, Saudi Arabia

7 **[Author note: confirm both affiliations; add the corresponding author and e-mail]**

8 *Running head:* Velocity errors and AVO misclassification

9 **ABSTRACT**

10 Amplitude-variation-with-offset (AVO) classification of the top of a CO₂-charged reservoir depends
11 on the P-wave velocities assigned to the caprock and the reservoir. We derive closed-form sensi-
12 tivities of the two-term Shuey intercept A and gradient B to relative P-velocity errors, including
13 the dependence of Poisson's ratio on V_P . We validate the symbolic implementation against an
14 independent derivation and apply it to a Draupne shale over CO₂-saturated Sognefjord sand model
15 of the Smeaheia storage site, northern North Sea. The baseline response is Class III ($A_0 = -0.214$,
16 $B_0 = -0.021$). The intercept stays negative for all errors within $\pm 10\%$; the gradient is near zero
17 and highly sensitive. Two-term Shuey reflectivity predicts an apparent Class III→IV transition at
18 a minimum error-vector magnitude of 4.44%. We recompute this threshold with exact Zoeppritz
19 reflectivity validated to 2×10^{-15} . The exact normal-incidence gradient raises the threshold to
20 5.58%, and a two-term least-squares fit over 0° – 30° raises it to 8.23%. Two-term Shuey therefore
21 understates the error needed for misclassification by about 20%, but the misclassification itself is
22 physical. A symbolic second-order expansion shows that the Shuey gradient contains no second-
23 order term; its bias is a third-order artifact of the Poisson's-ratio parameterisation. Common-mode
24 velocity errors do not cancel in B : with Poisson's ratio treated as a function of V_P , the first-order
25 caprock sensitivity changes sign. Published seismic-to-well V_P prediction errors in the Smeaheia
26 reservoir interval are 13%–17% of the mean log velocity. At that level, AVO class assignment at

this interface is uncertain unless the elastic model is independently constrained.

INTRODUCTION

Amplitude variation with offset (AVO) is routinely used to infer pore-fluid content from the angle dependence of P-wave reflectivity (Rutherford and Williams, 1989; Castagna and Backus, 1993; Castagna et al., 1998). The intercept–gradient crossplot and the classification of Rutherford and Williams (1989), extended by Castagna et al. (1998), give the interpreter a compact vocabulary. A Class III response (negative intercept, negative gradient; brightening with offset) is the classical signature of a low-impedance gas or CO₂ sand beneath shale. A Class IV response (negative intercept, positive gradient; dimming with offset) arises when a soft sand is capped by a lithology of relatively high S-wave velocity, so that the S-wave contrast reverses the sign of the gradient (Castagna et al., 1998).

Amplitude-preserving processing aims to retain relative amplitudes with angle, but quantitative interpretation still rests on an elastic model. That model is used to build fluid-substitution templates, to forward-model expected responses, and to decide whether an observed anomaly is consistent with CO₂. The P-wave velocities of the model are among its least certain inputs. They come from sonic logs that may not represent the seismic-scale interval, from seismic inversion, or from machine-learning property prediction. At the Smeaheia CO₂-storage prospect, the published multi-attribute-regression and probabilistic-neural-network predictions of V_P have validation RMS errors of 380.91 m/s and 461.3 m/s, respectively (Hema et al., 2027, their Table 12). Relative to the mean Sognefjord log velocities in wells 32/4-1 and 32/2-1, these errors equal 12.80%–16.95% [nb §9.7].

For CO₂-storage monitoring, a velocity-induced change of AVO class has a direct interpretational cost. If velocity errors turn a genuine Class III CO₂ response into an apparent Class IV, the interpreter may reject a correct CO₂ interpretation or misplace a plume boundary. The concern is sharpened at Smeaheia because the brine-sand reference response lies close to $B = 0$ (this study), so fluid discrimination relies on small gradient differences.

This paper asks a narrow, quantitative question: how large must relative errors in the caprock and reservoir P-velocities be before the AVO class of the Smeaheia top reservoir changes, and how

much of the answer depends on the reflectivity approximation used? We make five contributions:

1. a closed-form sensitivity analysis of the Shuey (1985) intercept and gradient to layer V_P errors, cross-validated against an independent symbolic derivation;
2. the total (chain-rule) sensitivity of the gradient when Poisson’s ratio is treated as a function of V_P , and a demonstration that the fixed-Poisson’s-ratio sensitivity is misleading;
3. a machine-precision exact-Zoeppritz (Zoeppritz, 1919) recomputation of the class-change threshold, which separates approximation artifacts from physical effects;
4. a quantification of how strongly the threshold depends on the angle aperture used to estimate the gradient;
5. a symbolic second-order expansion of the exact gradient that identifies the origin of the Shuey bias and places the quadratic approximation of Wang (1999) within the complete second-order result.

THEORY AND METHOD

Elastic model

The reference model (Table 1) represents the Draupne Formation shale caprock over Sognefjord Formation sandstone at approximately 1200 m true vertical depth subsea in the Smeaheia area (Fawad et al., 2021a,b; Hema et al., 2027). The CO_2 -sand properties were obtained by Gassmann (1951) fluid substitution of the brine-sand model [Author note: specify the CO_2 equation of state or property source, saturation model, and the pressure and temperature used; the source notebook gives $S_{\text{CO}_2} \approx 0.6$, $T \approx 40^\circ\text{C}$, $P_{\text{pore}} \approx 11 \text{ MPa}$]. We treat the model as fixed input.

As a consistency check, Gassmann’s relations leave the shear modulus unchanged, which predicts $V_S^{\text{CO}_2} = V_S^{\text{brine}}(\rho^{\text{brine}}/\rho^{\text{CO}_2})^{1/2} = 1183.5 \text{ m/s}$, compared with the model value of 1180 m/s [nb §9.7].

The active scenario, shale over CO_2 sand, has contrasts $\Delta V_P/\bar{V}_P = -0.2857$, $\Delta V_S/\bar{V}_S = -0.0168$, and $\Delta\rho/\bar{\rho} = -0.1461$ [nb §9.2]. These contrasts are not small, so the accuracy of weak-contrast reflectivity cannot be taken for granted.

Table 1. Reference elastic model (fixed input). Subscript 1 denotes the caprock and subscript 2 the reservoir.

Property	Caprock (Draupne shale)	Brine sand	CO ₂ sand
V_P (m/s)	2600	2200	1950
V_S (m/s)	1200	1150	1180
ρ (kg/m ³)	2350	2150	2030
Poisson's ratio σ	0.3647	—	0.2111

Two-term Shuey attributes

Following Shuey (1985) in the form given by Castagna and Backus (1993) and Wiggins et al. (1983), the intercept, the P-velocity contrast, and the gradient are

$$A = \frac{\rho_2 V_{P2} - \rho_1 V_{P1}}{\rho_2 V_{P2} + \rho_1 V_{P1}}, \quad C = \frac{V_{P2} - V_{P1}}{V_{P2} + V_{P1}}, \quad (1)$$

$$B = A \left[D - \frac{2(1+D)(1-2\bar{\sigma})}{1-\bar{\sigma}} \right] + \frac{\Delta\sigma}{(1-\bar{\sigma})^2}, \quad D = \left[1 + \frac{(\rho_2 - \rho_1)/(\rho_2 + \rho_1)}{(V_{P2} - V_{P1})/(V_{P2} + V_{P1})} \right]^{-1}, \quad (2)$$

where $\bar{\sigma} = (\sigma_1 + \sigma_2)/2$, $\Delta\sigma = \sigma_2 - \sigma_1$, and

$$\sigma_i = \frac{V_{Pi}^2 - 2V_{Si}^2}{2(V_{Pi}^2 - V_{Si}^2)}. \quad (3)$$

The intercept in equation 1 is the exact normal-incidence plane-wave reflection coefficient; our Zoeppritz implementation reproduces it to 5.6×10^{-17} [nb §9.1]. We classify responses with $A < 0$ as Class III when $B < 0$ and as Class IV when $B > 0$ (Castagna et al., 1998). The class-change boundary studied here is therefore $B = 0$.

Linear error propagation

Let $r_i = \delta V_{Pi}/V_{Pi}$ be the relative P-velocity error in layer i . Because A and C depend on V_P only through the ratio V_{P2}/V_{P1} , their first-order relative errors take the form

$$\frac{\delta X}{X} = k_X (r_1 - r_2), \quad X \in \{A, C\}, \quad (4)$$

so errors common to both layers (common-mode errors) cancel exactly. The same form holds for B *only* if $\bar{\sigma}$ and $\Delta\sigma$ are held fixed during differentiation. We call the corresponding coefficient k_B the

fixed- σ sensitivity. Physically, σ_i depends on V_{Pi} through equation 3. The total sensitivity is then

$$\frac{dB}{d \ln V_{Pi}} = \left. \frac{\partial B}{\partial \ln V_{Pi}} \right|_{\sigma} + \frac{\partial B}{\partial \sigma_i} \frac{\partial \sigma_i}{\partial \ln V_{Pi}}, \quad \frac{\partial \sigma_i}{\partial \ln V_{Pi}} = \frac{V_{Pi}^2 V_{Si}^2}{(V_{Pi}^2 - V_{Si}^2)^2}, \quad (5)$$

which is derived in Appendix B. The second term is not antisymmetric in the two layers. As a result, common-mode errors do not cancel in B .

Nonlinear error mapping

To go beyond first order, we re-evaluate equations 1–3 exactly at perturbed velocities, recomputing σ at each perturbed V_P and leaving V_S and ρ unperturbed. We use (1) one-dimensional slices (caprock only, reservoir only, and $r_2 = -r_1$), (2) the full error plane $(r_1, r_2) \in [-10\%, 10\%]^2$, and (3) radial spokes $r_1 = |r| \cos \phi$, $r_2 = |r| \sin \phi$ with $|r| \leq 10\%$. Along each spoke we locate $B = 0$ by bracketed root finding (Brent’s method, tolerance 10^{-12}). The minimum-radius threshold is then found by bounded minimisation over ϕ .

Exact Zoeppritz reflectivity and gradient estimators

We implemented the exact plane-wave P–P reflection coefficient twice, independently:

- by solving the 4×4 Zoeppritz system for $[R_{PP}, R_{PS}, T_{PP}, T_{PS}]$ at each angle;
- by the explicit closed form of Aki and Richards (1980, their equation 5.39) (Appendix A), written in vertical slownesses with a complex square-root branch so that it remains valid beyond critical angles.

An exact gradient is not uniquely defined, because exact R_{PP} is not linear in $\sin^2 \theta$. We therefore use two definitions:

- the *exact tangent gradient*, $B_Z^{\tan} = \partial R_{PP} / \partial (\sin^2 \theta)|_{\theta=0}$, obtained from a degree-6 least-squares polynomial in $\sin^2 \theta$ fitted to exact R_{PP} on 0° – 20° . This is the exact counterpart of the Shuey gradient, which is also a normal-incidence slope. Between polynomial degrees 6 and 10 it changes by 1.0×10^{-9} [nb §9.2];

- the *two-term least-squares gradient*, B_Z^{LSQ} : the slope of $R_{\text{PP}} \approx A + B \sin^2 \theta$ fitted to exact R_{PP} on 0° – 30° . This is the gradient that a two-term AVO inversion of noise-free, ideally processed gathers would recover.

We also evaluate the linearised gradient of Aki and Richards (1980),

$$B_{\text{AR}} = \frac{1}{2} \frac{\Delta V_{\text{P}}}{\bar{V}_{\text{P}}} - 2 \left(\frac{\bar{V}_{\text{S}}}{\bar{V}_{\text{P}}} \right)^2 \left(\frac{\Delta \rho}{\bar{\rho}} + 2 \frac{\Delta V_{\text{S}}}{\bar{V}_{\text{S}}} \right), \quad (6)$$

of which Shuey’s expression is a re-parameterisation that agrees to first order in the contrasts.

Second-order expansion of the gradient

To explain the differences between the gradient estimators, we expand them in the mean-referenced relative contrasts $a = \Delta V_{\text{P}}/\bar{V}_{\text{P}}$, $b = \Delta V_{\text{S}}/\bar{V}_{\text{S}}$, and $r = \Delta \rho/\bar{\rho}$, at fixed $k = \bar{V}_{\text{S}}/\bar{V}_{\text{P}}$. The layer values are $X_{1,2} = \bar{X}(1 \mp x/2)$, with x the relative contrast of X . A symbolic series expansion of the exact tangent gradient (Appendix A) gives

$$B_Z^{\text{tan}} = B_{\text{AR}} + \underbrace{a B_{\text{AR}}}_{\text{reference angle}} + \underbrace{k^3(r + 2b)^2}_{\text{shear}} \underbrace{- \frac{k}{4} r^2}_{\text{density}} + O(\chi^3), \quad (7)$$

where χ denotes the magnitude of the contrasts. The same expansion of the Shuey gradient (equation 2) gives

$$B_{\text{Shuey}} = B_{\text{AR}} + O(\chi^3), \quad (8)$$

that is, the Shuey gradient has no second-order term at all.

Wang (1999) derived a quadratic approximation of R_{PP} , his equation 10. In terms of the mean of the incidence and transmission angles, $\bar{\theta}$ (his equation 6), its only second-order term is $(\bar{\beta}/\bar{\alpha})^3 \cos \bar{\theta} \sin^2 \bar{\theta} (\Delta \rho/\bar{\rho} + 2\Delta \beta/\bar{\beta})^2$. The gradient of this term is exactly the shear term of equation 7. Wang obtains his approximation after assuming $(\Delta \rho/\bar{\rho})^2 \approx 0$ (his equation A-10), which removes the density term. The reference-angle term arises because the linearised formulas are written in $\bar{\theta}$, whereas the AVO gradient is measured against the incidence angle; since $\sin^2 \bar{\theta} \approx (\bar{V}_{\text{P}}/\bar{V}_{\text{P1}})^2 \sin^2 \theta$ at small angles, converting between the two multiplies the gradient by a positive factor. Equations 7 and 8 were derived with SymPy from the explicit solution of Appendix A and

from equation 2. They were verified numerically by scaling the Smeaheia contrasts by $s = 1$ to $1/16$: the residuals decrease as s^3 and converge to the symbolic third-order coefficients [nb §11.1–11.2].

Verification

The symbolic engine (SymPy) was checked against an independent Mathematica derivation using that derivation’s own test model. The Python implementation reproduces all six reference outputs to the printed precision (Table 2) [nb §2.1].

Table 2. Cross-validation of the symbolic engine against an independent Mathematica derivation. Test model: $V_{P1} = 3599$ m/s, $V_{P2} = 3930$ m/s, $V_{S1} = 2341.45$ m/s, $V_{S2} = 2304.5$ m/s, $\rho_1 = 2313.9$ kg/m³, $\rho_2 = 2388.72$ kg/m³.

Quantity	Python (SymPy)	Mathematica
A	+0.059832	+0.0598318
B	+0.041834	+0.0418341
C	+0.043963	+0.0439633
k_A	−8.32684827	−8.32685
k_B (fixed σ)	+24.87862148	+24.8786
k_C	−11.35113011	−11.3511

Both Zoeppritz implementations were required to agree to better than 10^{-13} before either was used. We tested four models on 0° – 60° : the two Smeaheia scenarios, the Mathematica test model, and a model with a P-wave critical angle of 41.8° . The maximum discrepancies are listed in Table 3 [nb §9.1]. Finally, all three normal-incidence gradients converge to $B_Z^{\tan}$ as the contrasts vanish, with relative deviation proportional to the contrast. For contrast parameters of 10^{-1} , 10^{-2} , and 10^{-3} the deviation is 9.5×10^{-2} , 1.3×10^{-2} , and 1.4×10^{-3} for B_{AR} , and 6.9×10^{-2} , 1.3×10^{-2} , and 1.4×10^{-3} for the Shuey gradient [nb §9.2].

Table 3. Verification of the exact Zoeppritz implementations (maximum absolute discrepancy over four models and 121 angles on 0° – 60°).

Check	Maximum discrepancy
4×4 matrix solution vs. explicit Aki and Richards (1980) form	1.86×10^{-15}
$R_{PP}(0)$ vs. $(Z_2 - Z_1)/(Z_2 + Z_1)$	1.73×10^{-16}
Energy-flux conservation, $ \sum E - 1 $	7.77×10^{-16}
PyLops <code>zoeppritz_pp</code> (pre-critical angles)	5.41×10^{-16}
bruges <code>zoeppritz_rpp</code> (conjugated post-critically ^a)	1.26×10^{-15}

^abruges uses the opposite Fourier sign convention; PyLops returns NaN beyond a critical angle.

RESULTS

Baseline response and linear sensitivities

The shale over CO₂-sand baseline gives $A_0 = -0.2137$, $B_0 = -0.0211$, and $C_0 = -0.1429$, a Class III response. The shale over brine-sand reference gives $A_0 = -0.1273$ and $B_0 = +0.0023$ [nb §3]. With $A < 0$ and $B > 0$, the brine reference therefore falls marginally within the Class IV quadrant. The linear coefficients at the CO₂ baseline are $k_A = +2.233$, $k_B = -31.277$ (fixed σ), and $k_C = +3.429$ [nb §3]. The large relative sensitivity of B is mainly a consequence of the small baseline gradient. The absolute fixed- σ sensitivity $\partial B / \partial \ln V_{P1} = +0.660$ is unremarkable, but it is divided by $|B_0| = 0.021$ [nb §9.6].

Nonlinear response over the error plane

The intercept is nearly linear in the velocity errors and remains negative over the entire $\pm 10\%$ error plane, with a maximum of $A = -0.116$ (Figure 1a) [nb §9.8]. The stack-amplitude bright spot is therefore robust to these errors. The gradient, in contrast, is strongly nonlinear (Figure 1b). The product $A \cdot B$, which is $+0.00451$ at the baseline, ranges from -0.00676 to $+0.01444$ over the error plane and changes sign across the $B = 0$ contour (Figure 2) [nb §5].

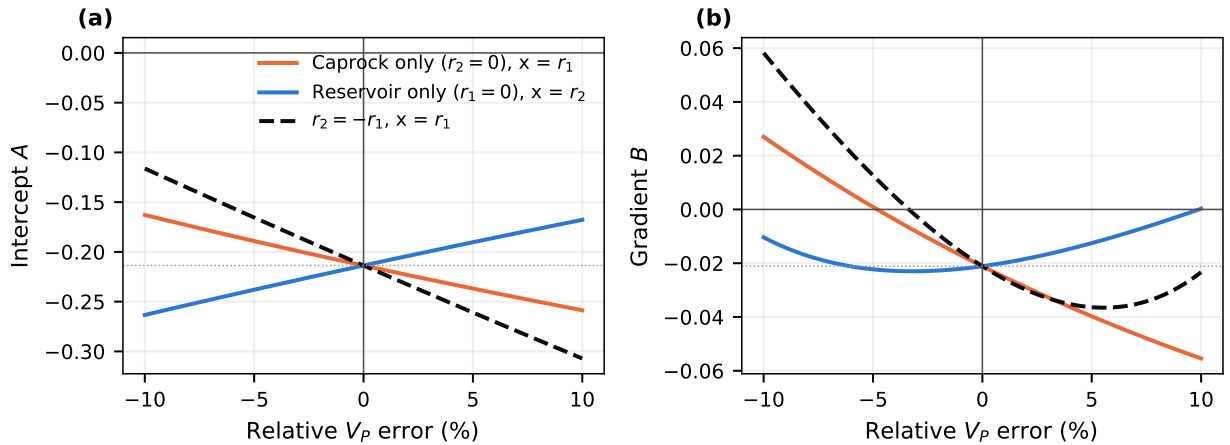


Figure 1. (a) Intercept A and (b) two-term Shuey gradient B versus relative V_P error for caprock-only ($r_2 = 0$), reservoir-only ($r_1 = 0$), and $r_2 = -r_1$ perturbations. Shale over CO₂ sand; σ is recomputed at each perturbed velocity. Dotted lines mark the baseline values.

Figure 3 shows the migration of the response in the crossplot. At $|r| = 10\%$ the Shuey gradient ranges from -0.0554 (caprock-only positive error, $\phi = 0^\circ$) to $+0.0304$ ($\phi = 135^\circ$, caprock slow and

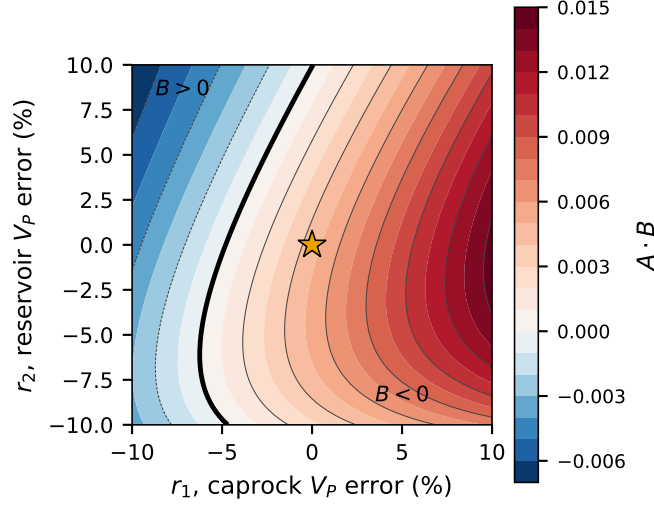


Figure 2. Product $A \cdot B$ over the (r_1, r_2) error plane (two-term Shuey). The thick black line is $B = 0$, the apparent Class III/IV boundary; the star marks the baseline. The colour scale is diverging about zero.

reservoir fast) [nb §9.3]. With two-term Shuey, $B = 0$ is crossed at $|r| = 4.752\%$ along $\phi = 135^\circ$, which corresponds to $r_1 - r_2 = -6.721\%$. The closest approach of the boundary to the baseline is $|r| = 4.444\%$ at $\phi = 156.6^\circ$, that is, $r_1 = -4.079\%$ and $r_2 = +1.765\%$ [nb §9.3]. The common-mode spokes ($\phi = 45^\circ, 225^\circ$) are not short. The $\phi = 225^\circ$ spoke reaches $B = +0.0033$ at $|r| = 10\%$ [nb §9.3], a first indication that common-mode errors do not cancel in B .

Exact Zoeppritz versus two-term Shuey

At the baseline the gradient estimators differ substantially (Figure 4, Table 4) [nb §9.2]. The two-term Shuey gradient (-0.0211) is 0.705 times the exact tangent gradient (-0.0299), that is, about 30% closer to the class boundary. The Aki-Richards gradient (-0.0445) errs in the opposite direction. The 0° – 30° least-squares gradient (-0.0513) is steeper than the tangent, because exact R_{PP} curves downward in $\sin^2 \theta$ at this interface (Figure 4).

Three results follow (Figure 5):

1. **The misclassification is physical.** Every exact-reflectivity estimator predicts a Class III→IV transition within $\pm 10\%$ when the caprock velocity is underestimated relative to the reservoir velocity.
2. **Two-term Shuey places the threshold too low.** Along $\phi = 135^\circ$ the exact tangent

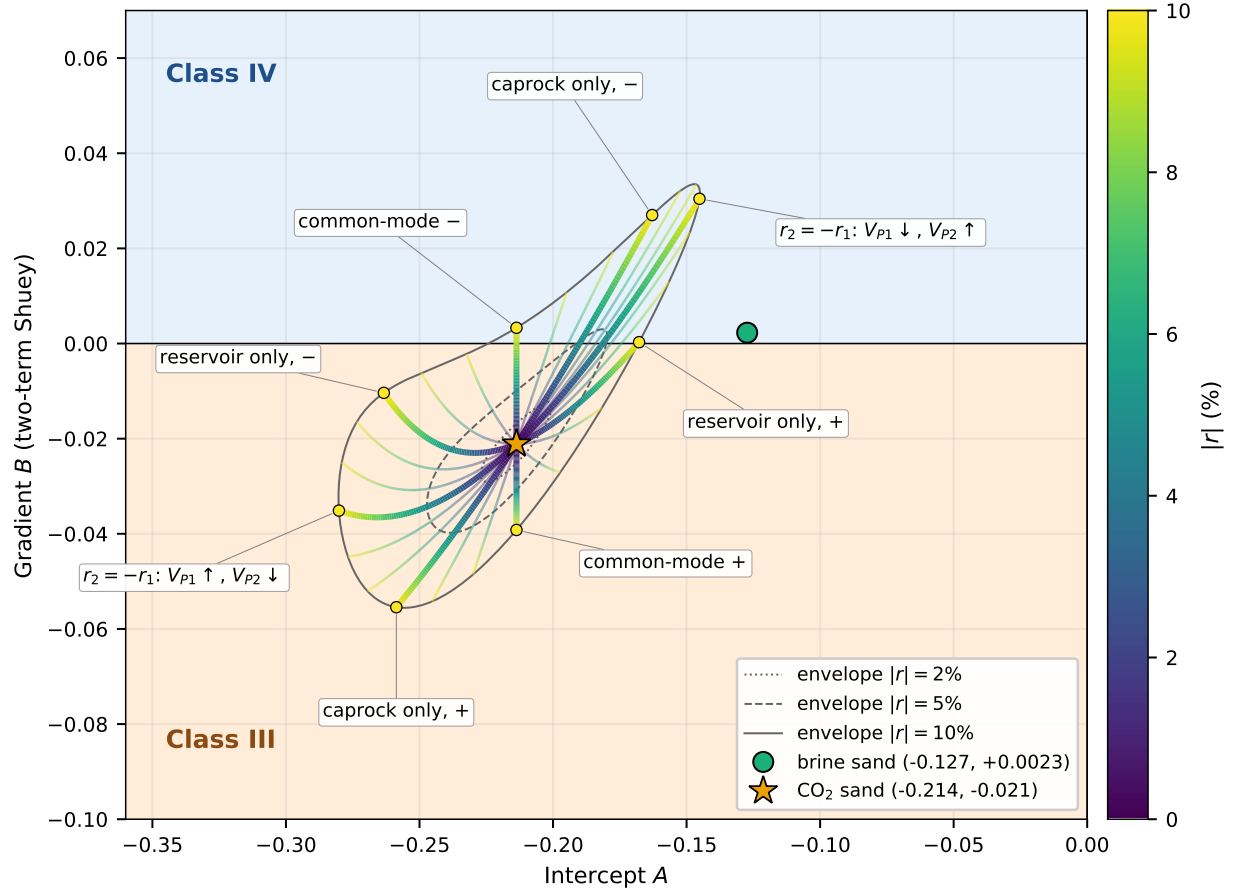


Figure 3. Radial error spokes in the A - B crossplot (two-term Shuey), coloured by error magnitude $|r|$, for 24 error directions ϕ in the (r_1, r_2) plane; the eight labelled spokes are the cardinal and diagonal directions. Envelopes are shown at $|r| = 2\%$, 5% , and 10% . The brine-sand reference is shown for comparison.

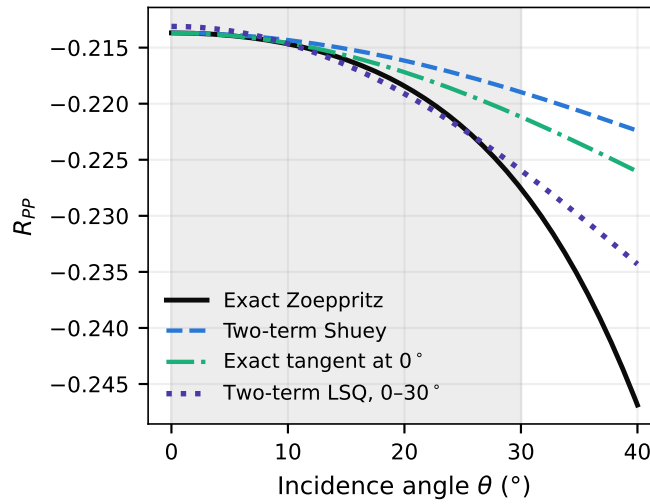


Figure 4. Exact Zoeppritz $R_{PP}(\theta)$ at the baseline (shale over CO_2 sand) compared with the two-term Shuey curve, the exact normal-incidence tangent, and the two-term least-squares fit over 0° - 30° (shaded).

Table 4. Baseline gradient and $B = 0$ threshold $|r|$ (%) by gradient estimator, along selected error directions ϕ and at the closest approach to the baseline; “–” denotes no crossing within $|r| \leq 10\%$. No estimator crosses $B = 0$ along $\phi = 0^\circ$, 45° , 270° , or 315° . The last column is the fraction of the $\pm 10\%$ error square in which $B > 0$ [nb §9.2, §9.4, §9.5].

Estimator	B_0	$\phi = 90^\circ$	$\phi = 135^\circ$	$\phi = 180^\circ$	$\phi = 225^\circ$	Min. $ r $ (ϕ)	$B > 0$
Two-term Shuey	−0.0211	9.900	4.752	4.802	8.826	4.444 (156.6°)	29.55%
Aki–Richards	−0.0445	–	6.206	7.064	–	6.035 (148.5°)	16.95%
Zoeppritz, tangent	−0.0299	–	5.755	6.484	–	5.580 (149.2°)	19.05%
Zoeppritz, LSQ 0–30°	−0.0513	–	8.455	9.628	–	8.225 (148.5°)	8.75%

threshold is 5.755% versus 4.752% for Shuey, a shortfall of 1.003 percentage points, or 17.4% of the exact value. At the closest-approach point the thresholds are 5.580% versus 4.444%, a shortfall of 1.136 points, or 20.4% [nb §9.4]. The bias comes mainly from the baseline gradient, which Shuey places too close to zero.

3. Some Shuey crossings are artifacts. Two-term Shuey predicts crossings for reservoir-only errors ($\phi = 90^\circ$, at 9.900%) and for common-mode errors ($\phi = 225^\circ$, at 8.826%). No exact estimator reproduces either crossing.

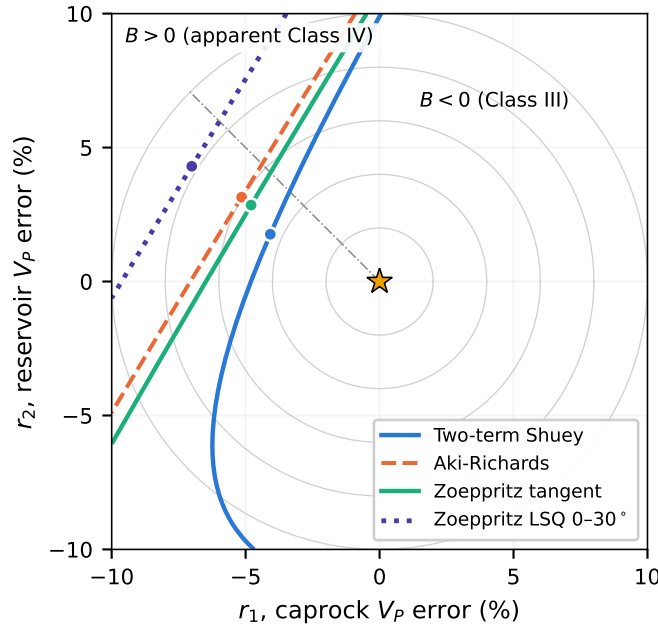


Figure 5. Class III/IV boundary ($B = 0$) in the (r_1, r_2) error plane for two-term Shuey, Aki–Richards, the exact Zoeppritz tangent, and the 0°–30° least-squares Zoeppritz gradient. Dots mark each boundary’s closest approach to the baseline (star); the dash-dotted line is the $\phi = 135^\circ$ spoke; grey circles are $|r| = 2\%$ – 10% .

Dependence on the angle aperture

When B is estimated by a two-term fit to exact reflectivity, both the baseline gradient and the threshold depend on the aperture (Table 5) [nb §9.4]. The threshold along $\phi = 135^\circ$ rises from 6.993% for a 0° – 20° fit to 9.355% for 0° – 35° . With a 0° – 40° fit, no crossing occurs within $\pm 10\%$.

Table 5. Aperture dependence of the two-term least-squares gradient fitted to exact Zoeppritz reflectivity [nb §9.4].

Aperture	B_0 (LSQ)	Threshold along $\phi = 135^\circ$
0° – 20°	−0.0389	6.993%
0° – 25°	−0.0443	7.662%
0° – 30°	−0.0513	8.455%
0° – 35°	−0.0601	9.355%
0° – 40°	−0.0711	—

Poisson’s ratio and the failure of common-mode cancellation

Evaluating equation 5 at the baseline gives the following [nb §9.6, §10.1]:

- *Fixed σ* : $\partial B / \partial \ln V_{P1} = +0.6603$ and $\partial B / \partial \ln V_{P2} = -0.6603$. The sum is zero, so common-mode errors cancel.
- *Total*: $\mathrm{d}B / \mathrm{d} \ln V_{P1} = -0.4050$ and $\mathrm{d}B / \mathrm{d} \ln V_{P2} = +0.1114$. The sum is -0.2936 , so common-mode errors do not cancel.

The chain-rule factors are $\partial B / \partial \sigma_1 = -3.0974$ and $\partial B / \partial \sigma_2 = +0.8467$, with $\partial \sigma_1 / \partial \ln V_{P1} = 0.3439$ and $\partial \sigma_2 / \partial \ln V_{P2} = 0.9115$. The caprock sensitivity therefore changes sign.

To first order along $\phi = 135^\circ$, the fixed- σ sensitivity predicts that B becomes more negative, with no crossing at all. The total sensitivity predicts a crossing at 5.782%, against 4.752% from the full nonlinear Shuey evaluation [nb §9.6].

Common-mode errors of $\pm 10\%$ also move B when it is computed from exact reflectivity. The exact tangent gradient changes from -0.0299 to -0.0433 ($+10\%$) and to -0.0117 (-10%), while A is exactly invariant [nb §9.6]. The reason is that exact R_{PP} depends only on velocity ratios and density ratios. Scaling V_P in both layers while holding V_S fixed changes V_S/V_P , and hence σ , in each layer.

Origin of the Shuey bias

Table 6 decomposes the two baseline gradients by order in the contrasts [nb §11.3]. The exact tangent gradient is almost entirely first plus second order. Aki–Richards (-0.04450) plus the complete second-order term ($+0.01455$) gives -0.02996 , against the exact -0.02994 . The largest second-order contribution is the reference-angle term ($+0.01272$), followed by the shear term of Wang (1999) ($+0.00462$) and the density term (-0.00279). The Shuey gradient has no second-order contribution. Its third-order term, however, is large ($+0.02163$) and overshoots the missing second-order correction. The net result is a gradient 0.00882 closer to $B = 0$ than the exact value. The bias of the Shuey gradient toward the class boundary is therefore a third-order artifact of its Poisson’s-ratio parameterisation, not an incomplete second-order correction.

Table 6. Decomposition of the baseline gradient by order in the relative contrasts ($a = -0.2857$, $b = -0.0168$, $r = -0.1461$, $k = 0.5231$) [nb §11.3].

Contribution	Exact tangent B_Z^{tan}	Two-term Shuey
First order (B_{AR})	-0.04450	-0.04450
Second order: reference angle, aB_{AR}	$+0.01272$	
Second order: shear, $k^3(r + 2b)^2$	$+0.00462$	0
Second order: density, $-kr^2/4$	-0.00279	
Third order	$+0.00037$	$+0.02163$
Fourth and higher orders	-0.00035	$+0.00176$
Total	-0.02994	-0.02111

The second-order gradients also reproduce the class boundary (Table 7) [nb §11.4]. The complete second-order gradient places the $B = 0$ threshold within 0.12 percentage points of the exact tangent value. Wang’s equation 10 places it 0.19–0.20 points lower. Because converting between $\bar{\theta}$ and the incidence angle multiplies Wang’s entire gradient by a positive factor, the choice of angle changes its baseline value (-0.03053 when converted to the incidence angle, -0.03988 when $\bar{\theta}$ is read as the incidence angle) but not its threshold. The remaining shortfall is due to the omitted density term.

Table 7. Baseline gradient and $B = 0$ threshold $|r|$ (%) for the second-order gradients, with the exact tangent and two-term Shuey values for reference [nb §11.4].

Estimator	B_0	$\phi = 135^\circ$	Min. $ r $ (ϕ)
Exact, to second order (equation 7)	−0.02996	5.876	5.697 (149.2°)
Wang (1999), equation 10, incidence angle	−0.03053	5.554	5.390 (149.0°)
Zoeppritz tangent (exact)	−0.02994	5.755	5.580 (149.2°)
Two-term Shuey	−0.02111	4.752	4.444 (156.6°)

DISCUSSION

What exact reflectivity changes

The conclusion drawn from two-term Shuey reflectivity survives exact reflectivity: differential V_P errors of a few percent can change the apparent AVO class at the Smeaheia top reservoir. Its magnitude does not survive unchanged. At the contrasts of this interface ($\Delta V_P/\bar{V}_P \approx -0.29$), the second-order correction to the gradient (+0.0146) is a third of the first-order value. Aki–Richards omits it, and Shuey replaces it with a larger third-order term, so the two bracket the exact tangent gradient from opposite sides (Tables 4 and 6). Two-term Shuey therefore overstates the vulnerability of the classification by 17%–20% in required error magnitude, and it produces spurious crossings for reservoir-only and common-mode errors. Where a closed-form gradient is preferred to the exact solution, equation 7 is the appropriate choice at contrasts of this size: it reproduces the exact gradient to 2×10^{-5} and the class threshold to about 0.1 percentage points. Wang’s (1999) equation 10 is nearly as accurate, provided its mean angle is converted to the incidence angle when a gradient value is reported.

Which gradient should define the class?

The tangent gradient is the correct reference for judging the Shuey formula itself. A two-term inversion of recorded gathers, however, returns an aperture-dependent slope. For this interface that slope is steeper than the tangent, so the practical threshold moves to 7.0%–9.4% for apertures of 20° – 35° , and beyond 10% for a 40° aperture (Table 5). A class label obtained from a two-term fit is therefore partly a property of the acquisition aperture and not only of the rocks. Where B is close to zero, we recommend reporting the aperture together with any AVO class assignment.

Is a $\pm 10\%$ velocity error realistic?

The Smeaheia seismic-to-well V_P predictions of Hema et al. (2027) have validation RMS errors of 12.80%–16.95% of the mean Sognefjord log velocity [nb §9.7]. These are errors of a property-prediction workflow, not of processing velocities, and they refer to the reservoir interval rather than to the caprock–reservoir differential. They nevertheless indicate that V_P uncertainties from several percent up to more than 10% are plausible for elastic models built at Smeaheia. At that level, the exact minimum thresholds of 5.6%–8.2% lie within the uncertainty.

Scope and limitations

The framework propagates errors in the layer P-velocities that enter the reflection coefficient, that is, errors in the elastic model used for interpretation. Processing-velocity errors act mainly through a different mechanism, the incidence angle assigned to each offset, which is not modelled here. Density and V_S errors are also not propagated. Because σ depends on V_S as well as V_P , a V_S error would interact with the effects described above.

The analysis further assumes isotropic, elastic, planar interfaces and a single interface without tuning, with noise-free reflectivity. Anisotropy, attenuation, thin-bed interference, and angle-dependent wavelet effects can all modify the observed gradient.

Model calibration

The brine-sand V_P of the reference model (2200 m/s) is 19.2% and 26.1% below the mean Sognefjord log velocities in wells 32/2-1 and 32/4-1. It is also below the minimum log value in both wells, which is 2436 m/s in 32/2-1 and 2748 m/s in 32/4-1 (Hema et al., 2027, their Tables 1 and 2) [nb §9.7]. The CO₂-sand model inherits this offset. The absolute values of A , B , and the thresholds reported here should therefore be read as representative of a soft-sand end member, not as calibrated Smeaheia values. The methodology and the relative conclusions (Shuey versus Zoeppritz, and the role of σ) do not depend on this calibration; the numerical thresholds do.

[Author note: Either justify the soft-sand end member (for example, from a seismic-scale or upscaled velocity at the injection point) or recalibrate the model to the well data and re-run the notebook before submission.]

CONCLUSIONS

A symbolic implementation of the Shuey intercept and gradient, cross-validated against an independent derivation, shows that the Class III baseline of the Smeaheia top reservoir has a robust intercept but a gradient close to zero. A Class III→IV misclassification is physically possible within $\pm 10\%$ P-velocity errors when the caprock velocity is underestimated relative to the reservoir.

The two-term Shuey threshold (minimum error magnitude 4.44%) is biased low. The exact tangent gradient gives 5.58%, and a 0° – 30° two-term fit to exact reflectivity gives 8.23%. The Shuey threshold is therefore 17%–20% below the exact tangent threshold. This offset is an approximation artifact; the crossing itself is not.

The origin of this bias is identified by a symbolic second-order expansion. The exact gradient equals the Aki–Richards gradient plus a reference-angle, a shear, and a density term; to second order this reproduces the exact gradient at Smeaheia to 2×10^{-5} . The Shuey gradient has no second-order term, and its proximity to the exact value comes from a third-order term that overshoots.

The fixed-Poisson’s-ratio sensitivity ($k_B = -31.28$) does not describe the physical response. Once σ is allowed to vary with V_P , common-mode V_P errors no longer cancel in B , and the first-order caprock sensitivity changes sign.

For CO₂-monitoring interpretation at interfaces where B is close to zero, AVO classes should be assigned with exact reflectivity and a stated angle aperture. Their robustness should also be tested against the velocity uncertainty of the elastic model.

ACKNOWLEDGMENTS

[Author note: Funding, data providers, and software acknowledgments.]

Use of generative AI. The authors used Claude (Anthropic), a large language model, to assist with (1) writing and verifying the Python code for the exact-Zoeppritz cross-check and the publication figures and (2) drafting and editing the manuscript text. The authors directed the analysis, checked every numerical result against the accompanying notebook, verified all references, and take full responsibility for the content. **[Author note: review and adjust this disclosure so that it accurately describes how AI was used throughout the project, as required by the SEG ethical guidelines]**

DATA AND MATERIALS AVAILABILITY

All computations, including every number, table, and figure in this paper, are contained in a single Jupyter notebook (Python with NumPy, SymPy, SciPy, and Matplotlib; independent checks with PyLops 2.8.0 and bruges 0.5.4). **[Author note: Archive the notebook with a DOI (for example, Zenodo) and cite it here.]** Well statistics are taken from Hema et al. (2027).

REFERENCES

- Aki, K., and P. G. Richards, 1980, Quantitative seismology: Theory and methods: W. H. Freeman and Company.
- Batzle, M., and Z. Wang, 1992, Seismic properties of pore fluids: *Geophysics*, **57**, 1396–1408, doi: 10.1190/1.1443207.
- Castagna, J. P., and M. M. Backus, eds., 1993, Offset-dependent reflectivity — Theory and practice of AVO analysis: SEG, *Investigations in Geophysics* 8.
- Castagna, J. P., H. W. Swan, and D. J. Foster, 1998, Framework for AVO gradient and intercept interpretation: *Geophysics*, **63**, 948–956, doi: 10.1190/1.1444406.
- Fawad, M., M. J. Rahman, and N. H. Mondol, 2021a, Seismic reservoir characterization of potential CO₂ storage reservoir sandstones in Smeaheia area, northern North Sea: *Journal of Petroleum Science and Engineering*, **205**, 108812, doi: 10.1016/j.petrol.2021.108812.
- Fawad, M., M. J. Rahman, and N. H. Mondol, 2021b, Seismic-derived geomechanical properties of potential CO₂ storage reservoir and cap rock in Smeaheia area, northern North Sea: *The Leading Edge*, **40**, 254–260, doi: 10.1190/tle40040254.1.
- Gassmann, F., 1951, Über die Elastizität poröser Medien: *Vierteljahrsschrift der Naturforschenden Gesellschaft in Zürich*, **96**, 1–23.
- Hema, G., S. P. Maurya, R. Kant, and A. P. Singh, 2027, Seismic inversion driven reservoir characterization and subsurface property prediction in the Smeaheia Field, northern North Sea: Implications for CO₂ storage: *Geoenergy Science and Engineering*, **268**, 214769, doi: 10.1016/j.geoen.2026.214769.
- Rutherford, S. R., and R. H. Williams, 1989, Amplitude-versus-offset variations in gas sands: *Geophysics*, **54**, 680–688, doi: 10.1190/1.1442696.

- 338 Shuey, R. T., 1985, A simplification of the Zoeppritz equations: *Geophysics*, **50**, 609–614, doi:
339 10.1190/1.1441936.
- 340 Wang, Y., 1999, Approximations to the Zoeppritz equations and their use in AVO analysis: *Geo-*
341 *physics*, **64**, 1920–1927, doi: 10.1190/1.1444698.
- 342 Wiggins, R., G. S. Kenny, and C. D. McClure, 1983, A method for determining and displaying the
343 shear-velocity reflectivities of a geologic formation: European Patent Application 0113944.
- 344 Zoeppritz, K., 1919, Erdbebenwellen VII. VIIb. Über Reflexion und Durchgang seismischer Wellen
345 durch Unstetigkeitsflächen: *Nachrichten von der Königlichen Gesellschaft der Wissenschaften zu*
346 *Göttingen, Mathematisch-Physikalische Klasse*, 66–84.

APPENDIX A

EXACT P-P REFLECTION COEFFICIENT

With ray parameter $p = \sin \theta / V_{P1}$ and vertical slownesses $q_{\alpha i} = (V_{Pi}^{-2} - p^2)^{1/2}$ and $q_{\beta i} = (V_{Si}^{-2} - p^2)^{1/2}$ (principal complex branch), define (Aki and Richards, 1980, equation 5.39)

$$a = \rho_2(1 - 2V_{S2}^2 p^2) - \rho_1(1 - 2V_{S1}^2 p^2), \quad b = \rho_2(1 - 2V_{S2}^2 p^2) + 2\rho_1 V_{S1}^2 p^2, \quad (\text{A-1})$$

$$c = \rho_1(1 - 2V_{S1}^2 p^2) + 2\rho_2 V_{S2}^2 p^2, \quad d = 2(\rho_2 V_{S2}^2 - \rho_1 V_{S1}^2), \quad (\text{A-2})$$

$$E = b q_{\alpha 1} + c q_{\alpha 2}, \quad F = b q_{\beta 1} + c q_{\beta 2}, \quad (\text{A-3})$$

$$G = a - d q_{\alpha 1} q_{\beta 2}, \quad H = a - d q_{\alpha 2} q_{\beta 1}, \quad (\text{A-4})$$

$$D = EF + GH p^2, \quad R_{PP} = \frac{(b q_{\alpha 1} - c q_{\alpha 2})F - (a + d q_{\alpha 1} q_{\beta 2})H p^2}{D}. \quad (\text{A-5})$$

At $p = 0$ this reduces to $R_{PP} = (\rho_2 V_{P2} - \rho_1 V_{P1}) / (\rho_2 V_{P2} + \rho_1 V_{P1})$, the intercept of equation 1.

APPENDIX B

TOTAL SENSITIVITY OF THE GRADIENT

Write the Shuey gradient (equation 2) as $B = B(V_{P1}, V_{P2}, \sigma_1, \sigma_2)$, holding V_S and ρ fixed. Because $\sigma_i = \sigma(V_{Pi}, V_{Si})$ (equation 3), the chain rule gives

$$\frac{dB}{d \ln V_{Pi}} = \left. \frac{\partial B}{\partial \ln V_{Pi}} \right|_{\sigma_1, \sigma_2} + \frac{\partial B}{\partial \sigma_i} \frac{\partial \sigma_i}{\partial \ln V_{Pi}}. \quad (\text{B-1})$$

With $x = V_{Pi}^2$ and $y = V_{Si}^2$, $\sigma_i = (x - 2y) / [2(x - y)]$, so $\partial \sigma_i / \partial x = y / [2(x - y)^2]$ and $\partial x / \partial \ln V_{Pi} = 2x$.

Hence

$$\frac{\partial \sigma_i}{\partial \ln V_{Pi}} = \frac{V_{Pi}^2 V_{Si}^2}{(V_{Pi}^2 - V_{Si}^2)^2} > 0. \quad (\text{B-2})$$

The first term of equation B-1 is antisymmetric between the layers, because B at fixed σ depends on V_P only through V_{P2}/V_{P1} . The second term is not antisymmetric, since $\partial B / \partial \sigma_1 \neq -\partial B / \partial \sigma_2$ in general. The first-order response to a common-mode error, $r_1 = r_2 = r$, is therefore $\delta B = r \sum_i (\partial B / \partial \sigma_i) (\partial \sigma_i / \partial \ln V_{Pi})$, which is not zero in general. Both equations were verified symbolically

370 and numerically in the accompanying notebook (§10.1).